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Executive Compensation Structures and Their Implications

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Abstract

The board of directors has invested a lot of energy in designing a compensation plan to align the interests of executives and shareholders. However, there is still no clear answer to a basic question in the literature: are the incentives stipulated in the contract consistent with the incentives that executives actually feel? Two related issues motivate this review. First, how does option-based compensation affect executive effort and risk-taking? Secondly, there is a difference between the cost of equity compensation on the company's books and the value that executives themselves perceive from these awards. Drawing on agency theory, contract theory and a large number of empirical studies, this article explores how stock options align management with the interests of shareholders, and how they lead to short-termism, earnings manipulation and excessive risk. In addition, this article also uses a large number of accounting literature to explain why published accounting data will mislead the board of directors and investors, causing them to misunderstand the true intensity of compensation contract incentives. Chapter 7 extends the analysis to the banking industry, pointing out that the unreasonable design of the remuneration system is one of the reasons for the 2007-2009 financial crisis, and the capital structure, deposit insurance and prudential supervision are inseparable from remuneration design. The core conclusion of the comment is that remuneration, accounting processing and supervision constitute an interrelated issue, not three independent issues.

Keywords: executive compensation, stock options, risk-taking, accounting costs, incentives, corporate governance, banking

1. Introduction

The operation of modern listed companies depends on the separation of ownership and control. Shareholders provide capital, but rarely participate in daily management. Managers make operational decisions that determine the value of the company, but they do not need to bear the

full cost of their own mistakes. Jensen and Meckling (1976) call this asymmetry an agency problem: managers may pursue goals that are contrary to the maximization of shareholder value, and shareholders cannot supervise the behavior of managers if they do not spend money. The executive compensation system aims to correct this problem. By linking the wealth of managers with the company's performance, at least in theory, managers should bear more decision-making consequences. However, whether this consistency is really realized in practice is an independent and far from settled question. Jensen and Murphy (1990) measured the sensitivity of executive wealth to the value of the company and found that the connection between the two was very weak. This result opened a 30-year study to explore the remuneration structure and whether it achieved the set goals. With the rise of stock options in the 1990s, this problem became more acute. By the beginning of the 21st century, options had become the largest single component of CEO compensation in the United States (Edmans, Gabaix and Jenter, 2017), and a series of corporate collapses, most notably Enron and WorldCom, have raised persistent concerns that option-intensive contracts may reward manipulation rather than real value creation. This concern still runs through the relevant literature.

The second less obvious problem is hidden under these concerns. The option valuation model used for accounting and information disclosure purposes assumes that investors are diversified and can trade freely to avoid potential risks. Executives are completely different from such investors. Their wealth and human capital are concentrated in a single company, and they are usually unable to hedge or sell unvested awards. Research by Hall and Murphy (2002) shows that executives who are risk-averse and do not diversify their valuation of options is much lower than their fair value accounting costs. Therefore, the remuneration cost reported by the company may be far from the incentive value actually perceived by executives. This mismatch will affect how the board of directors sets compensation benchmarks, how regulators interpret information disclosure, and how researchers measure the sensitivity of compensation and performance. The whole compensation evaluation system is based on a number that may not be accurate at all.

Existing work tends to treat these as two unrelated problems. One strand sits in corporate finance and behavioural agency research; another sits in accounting and valuation theory. Reviews that bring the two together are rare, even though a compensation package only makes sense as an effort-and-risk-inducing device once its accounting cost has been reconciled with what it actually

delivers to the executive holding it. This separation appears to have cost the field more than it usually gets credit for. Treating the two questions jointly lets this review speak to both how options shape behaviour and why the instruments boards actually observe, reported expense, vesting schedules, disclosed fair values, may be poor proxies for the incentives they are meant to measure. Whether anyone has checked this carefully before is a separate question, and largely an open one.

In the financial sector, both problems resurface, and arguably in a more consequential form. The leverage ratio of banks is abnormally high. Its funds mainly come from the equity that is on loan rather than risk-bearing, and the banks are interconnected. The collapse of a bank may cause systemic risks.

When bank remuneration encourages risky behavior, the ultimate price is not limited to shareholders; it will also affect depositors, creditors, and eventually even to taxpayers. The 2007-2009 financial crisis pushed this issue to the core of financial regulation. Policymakers in the United States, the United Kingdom and the European Union have taken countermeasures, including bonus deferral, malus and clawback provisions, and a compensation structure clearly linked to capital and liquidity regulations. These contents do not appear in the standard discussion of executive compensation of non-financial enterprises, so the banking industry occupies a chapter here, not just a brief mention.

This article aims to discuss the following two issues. First of all, we should review and critically evaluate the theoretical and empirical literature on how option remuneration affects the degree of effort of executives, risk-taking and the consequences of their unexpected behavior. Secondly, it is necessary to integrate the literature on the gap between the cost of remuneration accounting and the value that executives perceive from these awards, and explore the impact of this gap on the design and evaluation of remuneration contracts.

By examining these two issues side by side rather than separation, we can have a more comprehensive understanding of executive compensation, which is not only about whether a tool has achieved the expected incentives in principle, but also about whether the accounting and reporting environment surrounding the tool can let the board of directors, shareholders and regulators see the incentives it actually generated. The evidence reviewed in this article shows

that the answer is usually negative.

This review is organized around the following questions: What role does option compensation play in shaping the degree of effort and risk-taking of executives? What unexpected consequences will this form of remuneration bring? Why is there a difference between the accounting-recognized compensation expense and the incentive value perceived by executives? How does this affect the measurement of the relationship between remuneration and performance? What do these two questions have for compensation design and corporate governance, especially in the banking industry?

For the sake of data access, this review mainly focuses on large listed companies. Among these enterprises, remuneration is the easiest to observe, the division of ownership and control is the clearest, and most empirical studies are mainly carried out in these enterprises. Chapter 7 jumps out of this overall framework and takes the banking industry as an example to discuss. In the banking industry, compensation-driven risk-bearing extends far more than shareholders, but also to depositors, creditors and public finance. Bank compensation is now directly constrained by prudential supervision, including capital and liquidity requirements, compulsory deferred payment and clawback rules, which do not exist in non-financial enterprises. The literature in the non-financial field does not fully elaborate on this.

The argument of this Master Thesis is divided into three main parts. Section 2 to 4 build the core theory and empirical arguments: agency and contract theory, how options actually affect behavior, and the difference between accounting costs and perceived value. Section 5 and 6 take a step back, weigh the significance of corporate governance, and conduct a stress test on the shortcomings of the existing literature. Section 7 to 9 explore the banking industry in depth, integrate the three themes into an overall perspective, and finally explore the practical actions that the board of directors and regulators may take.

2. Theoretical Foundations

2.1 Agency Theory and Executive Compensation

Agency theory remains the dominant framework for thinking about executive pay. Jensen and Meckling (1976) lay out the basic argument: managers do not own the firm, so their incentives to

increase its value are weaker than an owner's would be. Performance-based compensation is the theory's prescribed fix. Tie managerial wealth to firm value, and the conflict narrows, at least in principle. Eisenhardt (1989) groups this literature into two related streams: positivist agency theory, which looks for governance mechanisms that limit self-interested behaviour, and principal-agent theory, which models the trade-off between risk-sharing and incentive strength. Compensation contracts sit at the meeting point of the two.

The theory runs into a problem here. Firm performance depends on things a manager cannot control, the economy, competitors, new technology among them, and paying or punishing managers for these factors is not efficient; it rewards luck rather than skill. Bertrand and Mullainathan (2001) demonstrate this directly. Oil-company executives were paid more when oil prices rose, even though the executives did not cause that price rise, and the gap between real-world pay contracts and what the theory predicts turns out to be large.

The second research route raises a more difficult question: will the board of directors really try to set compensation based on the interests of shareholders? Bebchuk and Fried (2004) believe that the board of directors often does not do this, because executives can take advantage of weak corporate governance and the board of directors to influence their compensation, so high compensation reflects rent extraction behavior, not good contractual relationships. Edmans, Gabaix and Jenter (2017) argue that broader evidence shows that these two forces work at the same time. Part of the compensation difference comes from the design of the real incentive mechanism; part from the power of the management; and the existing literature has not yet determined which force dominates in a particular company.

2.2 Contract Theory: Balancing Incentives and Risk

The problem raised by the contract theory is more specific than the agency theory: how should the remuneration structure be constructed when the company cannot directly observe the level of managerial effort of the management?

Holmstrom (1979) put forward an answer, that is, the principle of information. According to this principle, any signal containing information about the degree of effort should be included in the remuneration contract. Therefore, the remuneration scheme that combines stock income, accounting income and other indicators should be better than a single indicator. However, there

are two problems that hinder the realization of this answer.

Many signals are noisy and reflect more luck than effort; and increasing the number of signals will only give executives more room to manipulate signals. Neither of these two problems has a perfect solution. Holmstrom and Milgrom (1987) proposed a simpler model, assuming that the results are normal and risk-preference criteria.

Under these assumptions, the best contract is linear, that is, fixed salary plus fixed performance share. There is a clear trade-off in this result: higher performance wages will make risk-averse managers face greater uncertainty, so managers will demand higher salaries to take this risk. In effect, firms buy risk tolerance from executives, and pay the price for it.

2.3 Option-Based Compensation as a Potential Solution

Stock options seem to be a natural solution to the problem of risk avoidance, at least at first glance. Options can only generate income when the stock price exceeds the exercise price, so executives can share the rising income, while the downside risk is limited to the value of unvested options. The rationality of options is based on two concepts: first, options should be able to reduce excessive caution, because they reward the rise more than the punishment for the fall; second, options link remuneration to the stock price, and the stock price is a forward-looking indicator of value creation.

The pricing model proposed by Black and Scholes (1973) later became the basis of accounting and information disclosure rules.

However, empirical research complicates this simple truth from two aspects. Options may not work as the theory expects: the study of Hall and Murphy (2002) shows that the valuation of options by undiversified and risk-averse executives is much lower than their market price; the study of Dittmann and Maug (2007) goes further. They use the standard agency model to find that optimal contracts should rely more on restrictive stocks and less on options, which is inconsistent with the number of options actually used by the company. This leads to an obvious question: if options are not envisaged by contract theory, why are they so common? Murphy (2002) put forward another explanation, which is based on accounting rather than incentive logic: for decades, accounting standards have not required companies to include options in expenses, so

companies can grant options without reporting any costs. Overall, these three studies point in the same direction: option compensation may come more from favorable accounting loopholes than deliberately designed incentive mechanisms.

3. Option-Based Compensation and Executive Incentives

3.1 How Stock Options Influence Effort and Risk-Taking

Stock options are meant to line up the interests of executives and shareholders by tying managerial wealth to firm performance. When the share price rises, executives benefit from exercising their options and, in theory, work harder as a result. Two channels do the work here. Through effort, executives gain from higher stock prices and so are more motivated to improve firm performance. Through risk-taking, because options pay off asymmetrically, executives capture the upside from a project while facing limited losses if it fails. Risk-averse managers might, in principle, get past the kind of excessive caution that leads them to avoid risky but positive-value projects, an aversion that may stem from having so much human capital tied up in one company. That is precisely the behaviour options are designed to encourage. Whether they succeed is another matter.

Underlying this reasoning sits an assumption that all risk-taking benefits the firm. It does not, and pretending otherwise is where the theory oversells itself. A pay structure that encourages risk-taking can just as easily push managers toward projects with uncertain payoffs that never benefit shareholders. The effectiveness of stock options therefore hinges less on how much risk-taking they trigger than on the quality of what follows from it.

3.2 Empirical Evidence on Executive Behaviour

The empirical literature increasingly suggests that the link between option incentives and risk-taking is not just a cross-sectional correlation in one setting. An early wave of studies establishes this basic association using different measures of corporate risk-taking. Sanders (2001), Datta, Iskandar-Datta and Raman (2001) and Coles, Daniel and Naveen (2006) all measure the same underlying mechanism, the convex, vega-driven risk exposure created by option holdings, and link it to risk-taking indicators such as acquisition strategy, investment policy or financing choices.

These three studies differ in the scope of the decisions they examine. Sanders (2001) studies the balance between ownership and options as competing incentives. Datta, Iskandar-Datta and Raman (2001) focus on acquisition behaviour. Coles, Daniel and Naveen (2006) take a broader view that combines investment and financing policy. This difference in scope strengthens rather than weakens the common conclusion. The same basic mechanism appears across several different types of corporate decision, not just one.

What this cross-sectional evidence cannot show is the direction of causality. Firms that already plan high-risk strategies may simply grant more options, regardless of any real incentive effect, so the association could run in reverse. Two later studies address this endogeneity problem directly. They use unrelated sources of identifying variation and reach the same conclusion. Hayes, Lemmon and Qiu (2012) treat the mandatory expensing of options under FAS 123R as a near-natural experiment. They find that firms that cut back on option grants after the rule change became more risk-averse. Shue and Townsend (2017) use quasi-random variation in multi-year compensation schedules and reach a similar result. Two methodologically unrelated identification strategies, one based on a regulatory shock and the other on scheduling quirks in compensation plans, converge on the same finding. This convergence is the strongest evidence in the literature that the relationship between options and risk-taking is causal rather than merely correlational.

Table 3.1: Selected empirical studies on stock options and executive risk-taking

Study	Methodology	Sample	Main findings	Limitations
Sanders (2001)	Cross-sectional regression	Large US public firms, CEOs	Stock ownership curbs risk-taking; option holdings encourage it	Correlational; cannot rule out reverse causality
Datta, Iskandar-Datta and Raman (2001)	Cross-sectional regression on acquisition choices	US acquiring firms	Option-compensated executives pursue higher-risk, higher-growth acquisitions	Same endogeneity concern; acquisitions are one narrow decision margin
Coles, Daniel and Naveen (2006)	Cross-sectional regression, vega/delta decomposition	Large US public firms	Higher vega linked to riskier investment and financing policy	Vega and delta are jointly determined with pay level; hard to isolate causal effect
Hayes, Lemmon and Qiu (2012)	Quasi-natural experiment (FAS 123R shock)	US public firms, before/after 2005	Firms that cut options after mandatory expensing became less risky	Shock hit all firms at once; hard to isolate firm-specific responses
Shue and Townsend (2017)	Quasi-random variation from multi-year option plans	US public firms	Confirms causal link between options and risk-taking	Randomness of plan timing is only quasi-exogenous

On effort incentives, the evidence is notably weaker, in part because effort itself is hard to observe. Hanlon, Rajgopal and Shevlin (2003) find that a dollar of option value tracks with roughly \$3.71 of future income growth, though the relationship is concave, meaning diminishing returns set in as option value rises.

3.3 Unintended Consequences of Option-Based Compensation

In addition to the expected effect, the option itself will also bring some problems. The literature on unexpected consequences mainly focuses on two related mechanisms: earnings manipulation and opportunistic timing of granting. Regarding earnings manipulation, Burns and Kedia (2006) and Bergstresser and Philippon (2006) studied the same potential incentive mechanism, that is, the value of options is highly dependent on stock prices, and came to the unanimous conclusion that executives with stronger option-related incentives are more likely to be related to financial

fraud and earnings management. The main difference between the two studies is the way the results are measured. One study uses fraud allegations and the other uses discretionary accrual items. This difference makes the consistency between the two more valuable than the results of a single study.

Regarding the timing of the grant, Yermack (1997) and Lie (2005) independently recorded the timing of certain options, which were carefully planned, around the announcements that were beneficial to the company, which actually traced back the buying price of executives. Heron and Lie (2007) expanded this research direction, showing that this model has weakened after the tightening of information disclosure requirements. In short, all three studies point to the same potential behavior: opportunism. This also shows that this behavior will respond to regulation: when the cost of exploiting loopholes rises due to the accelerated speed of mandatory disclosure, loopholes are often blocked. This provides relatively encouraging evidence in the literature. Targeted supervision can reduce opportunistic behavior without complete elimination of choice.

Short-termism is the third related problem. Executives may push up short-term stock prices while cutting investments that will bring returns in the long run. Edmans, Fang and Lewellen (2017) found that managers who are about to acquire a large equity are more likely to cut investments and manipulate earnings simply to meet market expectations. This finding clarifies a broader view: even a contract that successfully aligns the interests of managers and shareholders in the long run may still generate short-term distortion around the vesting date.

3.4 Critical Evaluation

In general, options affect behavior, although they do not always develop in the expected direction. They may prompt executives to manipulate performance and focus only on short-term interests rather than hard work and invest wisely, and this difference largely depends on the specific situation. Options may be very suitable for young technology companies, because the benefits of innovation are huge, and the cost of risk avoidance is especially high. However, the same structure may cause real damage in mature companies with stable cash flow, or in banks where risk-taking will bring systemic consequences. A study by Siegel and Hambrick (2005) shows that pay differentials have a greater impact on the performance of technology-intensive companies, which reminds us that the same compensation mechanism may have different effects

in different industries. The mixed use of wages, options, restricted stocks and bonuses is a standard practice, but it often makes executives work in a contradictory direction: the high volatility (vega) brought by options will encourage them to work in one direction, while a large number of restricted stock positions will motivate them to work in another. Devers, Wiseman and Holmes (2007) found that executives attach a kind of endowment value to the options they already hold. With the accumulation of real-value options, their risk aversion will also increase, and the standard incentive model cannot capture this dynamic. These conflicting incentive mechanisms, rather than any single incentive tool, may explain many inconsistencies in empirical research.

4. The Mismatch Between Accounting Costs and Actual Incentives

4.1 Accounting Treatment of Executive Compensation

Before December 2004, the U.S. Accounting Standards did not include par value stock options in the reported earnings, which was based on APB Opinion No. 25, which gave the company a good reason to pay executive compensation with options, because these compensations were never reflected as costs on the books. The traditional method to value employee stock option plans was the intrinsic value of the options. Since they were issued at the money, the accounting cost of the option plans was zero on the grant day. Murphy (2002) believes that it was this accounting standard, not any incentive mechanism, that drove the options boom in the 1990s.

In 2005, the FAS 123R changed the situation: companies now have to charge options at fair value on the grant date. The International Accounting Standard (IASB) issued the IFRS 2 (International Financial Reporting Standard), which follows the same path as the US standard: stock option plans were to be valued using the fair value. Both accounting standards said that the fair value is the value that the stock option plans would have if they were traded in a market. Then, that valuation must be based on accepted financial principles such as the time value of money, expectations and risk-neutral valuation, like in the Black-Scholes framework. However, the accounting standards also recognize that stock option plans show some differences with respect to standard financial options. Then, standard valuation methods like the Black-Scholes formula may be not suitable and numerical methods like binomial trees and simulations should be used. The main differences between stock option plans and financial traded options are: i) The

vesting period, where the exercise of the options is not allowed; ii) The forfeiture risk, if the executive/employee is fired or leave the firm voluntarily, unvested options are cancelled and vested options must be exercised immediately, independently of the moneyness; iii) The early-exercise behaviour, stock option holders use to exercise their options earlier than they would be exercised by a risk-neutral agent. As we mention, this phenomenon may be due to the lack of diversification of the executive, since her salary, bonuses, and her human capital are invested in the firm.

All these features were included in model of Hull and White (2004) which is based on a binomial tree. León and Vaello-Sebastià (2009) propose a method based on Monte Carlo simulations which allow to incorporate several state-variables, such as stochastic volatility and/or time-varying interest rates, besides the previous features.

Hayes, Lemmon and Qiu (2012) recorded what happened next. The use of options has declined sharply, replaced by restricted stocks, and this change is very rapid, spreading to companies of all sizes. This change in time is of great significance, indicating that it is not the incentive mechanism that has always supported the options, but the favorable accounting treatment.

Fair value accounting improves transparency, but does not solve all measurement problems, because option valuation still needs to make assumptions about expected volatility, dividend yield, risk-free interest rates and life expectancy. The company has considerable autonomy over these input parameters, and two studies show how the company uses this autonomy. Bartov, Mohanram and Nissim (2007) found that some companies use this autonomy to underestimate the value of options, especially some companies have weak governance. Choudhary (2011) further expands this finding and shows that this undervaluation becomes more common after compulsory expensing. Overall, the two studies came to a consistent conclusion: the company did not simply accept the new and higher reporting costs, but adjusted its reporting method to mitigate its impact.

4.2 Why Accounting Costs and Executive Value Differ

The valuation of remuneration by executives is different from that of shareholders and accountants. Hall and Murphy (2002) explained the reason: executives' portfolios are relatively single, and most of their wealth and careers are concentrated in one company. Therefore, for

executives in this situation, the value of high-risk stock options is far lower than the value for diversified external investors. Therefore, an option purchased by a shareholder for one dollar may be less than one dollar for the executive who obtains the option.

A related but different gap comes from three other factors, and they all point in the same direction. Dahiya and Yermack (2008) found that retirement restrictions reduce the valuation of executives' equity incentives; while Carpenter (1998) and the studies of Bettis, Bizjak and Lemmon (2005) show that executives exercise options earlier than predicted by the standard model. Early exercise, like retirement restrictions, will reduce the actual value obtained by executives from rewards, making it lower than the value assumed by the accounting model. All three studies came to the same conclusion: the standard valuation model overestimates the real value of executives' compensation.

In other words, accounting costs on the books do not always reflect the actual incentive effect. A compensation plan may look expensive on the books, but it has little incentive for the holder; vice versa, because option backdating or favorable timing can increase the actual income of executives, which will not be reflected in the reported expenses. If researchers only use accounting data as an indicator to measure the intensity of incentives, the data they use may be misleading.

4.3 Implications for Compensation Design

For compensation design, the gap between cost and value is crucial in several specific aspects. If the board of directors evaluates the compensation plan based on accounting costs rather than the actual perceived value of executives, it will overestimate the incentive effect for every dollar spent; and if executives significantly discount options, even the plan that looks expensive on the books may lack incentive. Before the introduction of the FAS 123R guideline, the board of directors issued options more frequently than when the actual economic cost was reflected in the income. The later shift to restricted stocks showed some intriguing phenomena: many companies have actually been subsidizing option grants at the expense of shareholder interests.

Since 2005, performance-based equity incentives have become more common, and their rewards can only vest after certain accounting or market goals are achieved. These tools do solve some of the problems of ordinary options by linking costs to specific results more directly. However,

Bettis et al. (2018) found that performance-based equity incentives are more difficult to value than ordinary options, because their returns depend on the nonlinear function of future performance indicators. The board of directors ultimately relies on the Monte Carlo simulation, and shareholders can hardly verify the assumptions of these simulations, let alone question them.

4.4 Critical Discussion of the Cost-Incentive Mismatch

In the relevant literature, the mismatch between accounting costs and incentives deserves more attention. Most of the research in this field is exploring whether executive compensation is too high; but perhaps the more valuable question is whether compensation can produce expected behavior, because high remuneration costs do not necessarily mean strong incentives, and low cost does not necessarily mean weak incentives.

The way of wage measurement exacerbates this problem. Researchers usually use published compensation values to study the relationship between compensation and performance, and once the accounting values deviate from the value that executives themselves perceive, the estimate of incentive intensity will become unreliable. This can explain why the research results are always contradictory but not completely satisfactory.

When evaluating executive compensation, shareholders and regulators rely heavily on accounting disclosures. If these disclosures cannot accurately reflect the incentive effect, the supervision mechanism based on these disclosures cannot do so. This seems to be the core of the whole battle for compensation measurement. If there is an error in the accounting treatment, all subsequent conclusions about the remuneration effect will be based on the wrong numbers.

5. Implications for Compensation Design and Corporate Governance

5.1 Implications for Board Decisions

For the remuneration committee of the board of directors, the above review contains a number of practical experiences, the most important of which is that the incentive structure is more important than the remuneration level. The focus of public discussion is often on the income of executives. However, theoretical and empirical studies continue to point out other aspects: the importance of the remuneration structure is at least as good as the size of the compensation, and

well-designed but smaller plans are often better than poorly structured but larger programs.

Among the design levers that the board of directors can use, the equity granting period may be the most important. A shorter equity grant period will generate strong short-term incentives, but it is also easy to lead to short-termism. Two studies point out the same solution from different perspectives. Research by Edmans, Fang and Lewellen (2017) shows that a longer period of equity granting can better align executives with the long-term value of the company; while the study of Edmans et al. (2012) goes further, believing that the best contract should allow executives to benefit from the company's performance even after retirement to prevent them from being manipulated at the end of their careers. Both studies came to the same conclusion that the compensation risk period should be extended, but few companies have really adopted this suggestion.

The benchmark analysis of the board of directors should also consider the cost-value gap. Market comparisons are usually based on accounting costs, but if the management of peer companies also discount options significantly, the actual remuneration market value of these peer companies may be closer than the accounting data shows. Ignoring this point, the benchmark analysis will only push up the nominal salary, but will not increase the incentive accordingly.

5.2 Shareholder Monitoring and Regulatory Responses

Shareholders now have a formal voice on compensation through say-on-pay voting, introduced in the UK in 2002 and in the United States in 2011 under Dodd-Frank. Two studies find that the mechanism works roughly as intended. In the UK, say-on-pay was followed by higher pay-performance sensitivity, especially at firms with high excess pay and weak prior performance (Ferri and Maber, 2013), and across 38 countries, mandatory say-on-pay is linked to a seven percent drop in CEO pay along with stronger pay-performance links (Correa and Lel, 2016). A third study complicates this picture. Kronlund and Sandy (2016) find that say-on-pay may simply push firms to expand less visible parts of pay, such as pensions, while holding down more visible parts, such as base salary. Taken together, these three studies agree that say-on-pay changes behaviour, but they do not agree on whether it truly strengthens the link between pay and performance or whether it simply moves pay into less visible forms.

Institutional investors have taken on a more active monitoring role too. Hartzell and Starks (2003)

find that concentrated institutional ownership goes along with both lower total CEO pay and stronger pay-performance sensitivity, consistent with the idea that concentrated investors have both the motive and the expertise to watch pay-setting closely. Accounting standards, FAS 123R especially, appear to have shaped pay structure more directly and more quickly than any voting-based mechanism has managed. The fast, near-universal shift away from options after 2005 suggests something the voting-rights literature cannot quite match: accounting treatment, not shareholder voice, was the primary driver of instrument choice.

5.3 Balancing Incentives, Simplicity, and Transparency

There is a real trade-off between the complexity and transparency of incentive mechanism design. Targeted contracts can solve specific problems. For example, linking equity grants to relative performance can eliminate the impact of luck, but the increased complexity makes it more difficult for shareholders to assess compensation, executives are easier to manipulate, and more dependent on independent consultants who are not always clear.

Relative performance evaluation is a good case to test this trade-off. Gong, Li and Shin (2011) found that nearly 25% of the companies in the S&P 1500 index component stocks have used this evaluation method in some form. In theory, the tool is attractive because it eliminates the impact of all companies in the industry at the same time; but in practice, it depends entirely on how the company chooses its peer group. Faulkender and Yang (2010) pointed out a possible problem: some companies choose peer companies with unusually high salaries just to justify paying higher salaries to their own executives.

Overall, these two studies expound the same point of view from opposite perspectives. The tool can only work when the peer selection process is transparent, and the lack of this transparency turns a mechanism meant to eliminate noise into a new way to raise wages.

No contract design can be applied to all situations. A reasonable combination of fixed remuneration, options, restricted shares and performance rewards depends on the company's characteristics, industry dynamics, the quality of corporate governance, and the behavior that the board of directors wants to encourage, and the standardized structure sometimes pushed by agent consultants is often not applicable to companies with unusual operating models.

No model can be perfectly applied to any company. Vesting design, institutional supervision and accounting-driven tool selection changes are all useful tools, but they are not enough to solve the problem in themselves. The banking industry clearly shows this limitation: capital requirements, liquidity supervision, and mandatory deferral and clawback provisions have an impact on the remuneration structure of banks, which cannot be reflected in non-financial companies at all.

6. Critical Review of the Empirical Literature

6.1 Competing Perspectives on Executive Compensation

Roughly speaking, empirical research literature can be divided into two camps. One faction believes that the actual observed remuneration contract is close to the effective result; the other faction believes that executives take advantage of the shortcomings of corporate governance to extract rents that exceed the expected range of effective contracts.

Some research results support the view of the effective contract camp: remuneration is related to the size of the company, remuneration is linked to performance, and the remuneration setting of some institutions seems to be more reasonable. Gabaix and Landier (2008) pointed out that the six-fold increase in CEO remuneration between 1980 and 2003 can be almost entirely attributed to the growth in the size of listed companies and the resulting increase in the marginal value of management talents. From this point of view, the high remuneration of executives reflects real scarce rent, not management power.

The rent-extraction camp also has its own research results, but they all revolve around a common theme: remuneration is not always linked to value creation. Research by Bertrand and Mullainathan (2001) shows that executive compensation depends more on luck than ability. Research by Garvey and Milbourn (2006) shows that pay is asymmetrical, and good luck is rewarded, while bad luck is not punished. A study by Heron and Lie (2007) shows that the timing of some options is carefully designed to profit from the favorable fluctuations of stock prices. Research by Core, Holthausen and Larcker (1999) shows that companies with weaker corporate governance tend to pay higher. These four studies differ in methods and backgrounds, but they all point in the same direction: they all show that compensation changes are not based on real performance appraisal, but opportunism disguised as an incentive mechanism.

Neither side has fully won the debate, and both sides have problems that the other side can take advantage of. Effective contract theory is difficult to explain why the sensitivity of remuneration and performance is often lower than the prediction of the standard model, nor can it explain why some compensation components seem to be aimed at hiding value rather than creating value. The rent-extraction theory itself also has shortcomings: it is difficult to explain the sharp increase in CEO remuneration even in companies with perfect corporate governance, and it is also difficult to provide transnational evidence consistent with the optimal contract forecast (Fernandes et al., 2013). The most likely explanation is that these two forces work at the same time, and their respective weights are constantly changing in different companies and governance systems. This may be a real feature of the remuneration setting mechanism, rather than a puzzle that requires more data to solve.

6.2 Methodological Challenges in Existing Research

Endogeneity is the most difficult problem faced by researchers here. The remuneration plan is not issued randomly, but reflects the characteristics of the company, the decision-making of the board of directors and the bargaining power of the management, all of which will directly affect the company's performance. Therefore, without a reliable identification strategy, it is difficult to interpret the correlation between the remuneration structure and the company's performance as a causal relationship.

Two studies have provided a partial solution, which uses different sources of variation that are more difficult to explain. Hayes, Lemmon and Qiu (2012) regarded the 2005 accounting standard change FAS 123R as an impact beyond the control of any single company, and found that companies that were forced to cut options would also reduce risks. Shue and Townsend (2017) used quasi-random time in the multi-year pay cycle to obtain similar results. Both studies are closer to real causal evidence, but each study has its limitations: the FAS 123R impact affects all companies at the same time, so it is difficult to separate the specific reaction of the company; and the compensation cycle is only quasi-random, because the company may arrange its plans in a way that is not completely random.

A second measurement problem sits underneath the first. Much of this literature uses Black-Scholes values as a stand-in for incentive strength, yet as Chapter 4 argued, these figures likely

diverge from how executives actually value their own pay. Using accounting cost as a proxy for incentive strength can understate the true relationship simply through noise in the measure. Tosi et al. (2000) illustrate the scale of this problem in a meta-analysis, finding that firm size explains more than 40 percent of the variance in CEO pay while performance explains less than 5 percent. Part of this gap reflects a real weak link between pay and performance; part of it is simply measurement error.

6.3 Contradictions in the Literature

There are several contradictions in these evidences that cannot be resolved. Jensen and Murphy (1990) believe that the motivation of CEOs is insufficient, but Hall and Liebman (1998) use more complete data to draw the opposite conclusion. Part of the difference is mainly due to the standard of measurement, because once the equity holdings other than the annual remuneration are included, the incentive seems to be much stronger.

The second contradiction is related to risk. Three studies examined the same basic problem, but came up with three different answers. Aggarwal and Samwick (1999) found that as corporate risks increased, the sensitivity of remuneration and performance decreased, which was very consistent with the standard entrusted agency theory. Prendergast (2002) put forward the opposite point of view from a theoretical perspective: in a risky environment, supervision is more difficult, agents need stronger motivation, not weaker motivation. Coles, Daniel and Naveen (2006) find a positive link between risk and vega, a result that fits neither of the first two stories cleanly. The three findings do not average out to a middle answer; instead they suggest that risk and compensation interact in more than one way, and that no single theory has yet captured all of them. A sharper account of how different kinds of risk interact with different pay instruments is still needed.

6.4 Research Gaps and Future Opportunities

Most research looks at CEOs of large U.S. public firms, leaving private firms, smaller public firms, and non-U.S. markets underrepresented. Institutional environments differ enormously, and compensation may play a different role in countries with concentrated family ownership than in the widely held firms typical of the United States. Fernandes et al. (2013) find that controlling

for governance and ownership structure narrows the apparent CEO pay gap between the United States and other developed countries, a reminder that much of the "pay gap" story is really a governance-and-ownership story in a different guise.

Below the CEO level, executives remain seriously understudied. Aggarwal and Samwick (2003) find that pay-performance sensitivity varies significantly by hierarchical level within firms, yet this variation rarely enters compensation theory, an odd omission given how much corporate strategy now relies on distributed teams.

Long-run effects are a further gap. Most studies measure outcomes over one to three years after a compensation change, while the dynamic contracting models reviewed by Edmans, Gabaix and Jenter (2017) predict that incentive effects build up and interact over time in ways short-window studies cannot capture. This question remains largely unanswered in the existing literature.

7. Executive Compensation and Risk-Taking in the Banking Industry

7.1 Why Banking Deserves Separate Treatment

In principle, the arguments built up so far apply to firms in any sector. Banking, though, is different, and not in a minor or incidental way. The most basic difference is capital structure. Non-financial firms typically finance around a third of their assets with debt, while commercial banks finance around ninety percent of theirs with debt, most of it in the form of deposits (Bolton, Mehran and Shapiro, 2015). Bank equity sits as a thin cushion beneath a huge base of fixed, largely insured claims, and that structure makes it behave something like a call option on the value of the bank's assets: value rises with asset volatility, while losses beyond the thin equity cushion fall on depositors, the deposit insurer, and, in a systemic shock, the taxpayer. This financing structure directly affects the incentive mechanism for management to take risks, not just as a background environment. John and John (1993) show that, in a general corporate-finance setting, the optimal sensitivity of management compensation to firm performance depends on how a company is financed.

Sensitivity depends on the financing method of the company, because high-leverage companies should theoretically provide lower remuneration-performance sensitivity to management, because a stronger incentive mechanism will interact with leverage and transfer risks to fixed

creditors. Banks are an extreme example of this logic. John, Saunders and Senbet (2000) pointed out that because deposits are insured, depositors have little reason to price or monitor bank risks. Therefore, the constraining effect of high-risk creditors on management to take risks has been weakened or disappeared. Therefore, executive compensation that rewards performance growth, such as stock options, may make already difficult problems worse rather than solving them.

There are three significant differences between banks and non-financial companies, and each difference is crucial. The standard agency theory believes that remuneration aligns managers with the interests of shareholders, but this does not apply to the banking industry, because once losses can be passed on to insured depositors and eventually to the public purse, shareholders themselves may prefer more risk than is socially optimal (Bebchuk and Spamann, 2010).

The operation mode of banks is also relatively opaque and interrelated, which makes supervision difficult, and the collapse of a bank may be a source of systemic risk; the externality of bank risk-bearing is greater and more imperceptible here than in almost any other field. Because of these externalities, bank remuneration is directly affected by factors such as regulation, capital requirements, liquidity rules and remuneration standards, which does not exist in non-financial enterprises. The remuneration design of the banking industry cannot be analyzed without its regulatory framework.

7.2 Deposit Insurance, Capital Structure and the Design of Bank Compensation Contracts

John, Saunders and Senbet (2000) regard deposit insurance as a source of moral hazard: because the profits from risk-taking belong to shareholders, and part of the losses are borne by insurers, banks have an inherent incentive to take too many risks. They pointed out that compensation can be offset by making the management's wealth sensitive enough to downside risks, because a flatter and less convex compensation structure can reduce risk transfer. From this point of view, remuneration supervision is a supplement to capital requirements, not a substitute. Bolton, Mehran and Shapiro (2015) extended this logic through a formal model involving shareholders, creditors, depositors and executives, showing that linking remuneration to market-based default risk measurement indicators (such as banks' credit default swap spreads) can curb excessive risk-taking behavior. However, shareholders may not be able or unwilling to sign such contracts because of the distortion caused by deposit insurance and unobservable tail risks. An important

inference of their model is that better bank governance can completely reduce the demand for incentive compensation, while more fierce competition for executives in the labor market will push up equity and risk-linked compensation. These two forces are in opposite directions, and which force dominates in any bank is an empirical question, not a theoretical one. Kolm, Laux and Loranth (2017) found that compensation supervision is significantly effective in preventing banks from overinvesting in risk-increasing strategies, but not effective in preventing underinvestment in risk-reducing strategies, which means that regulators should combine compensation supervision with capital supervision rather than relying on any of them alone. Hilscher, Landskroner and Raviv (2021) built a similar interaction model and came to roughly the same conclusion that compensation-based tools are best combined with capital-based prudential tools rather than replacing the latter.

In addition to the sensitivity to performance, there is another problem, that is, the maturity of the remuneration itself. Edmans and Liu (2010) proposed that internal debt, that is, unsecured, debt-like compensation, such as deferred remuneration and fixed income pensions, makes executives consistent with creditors rather than just shareholder interests, thus reducing the motivation to transfer risks to creditors. Sundaram and Yermack (2007) found that although internal debt has received far less attention from academia and regulators than equity compensation, in fact, the internal debt of many CEOs is already quite considerable. The right response to deposit insurance, in other words, is probably not to weaken incentive pay across the board, but to recompose it: less convexity, a longer time horizon, and part of it tied to instruments sensitive to downside risk.

7.3 Competition, Deregulation and the Level and Structure of Bank Pay

Competition and deregulation shaped bank CEO pay through two distinct channels documented in the literature, pay-performance sensitivity and the choice of pay instruments.

On pay-performance sensitivity, several studies of the deregulation of U.S. interstate banking in the 1980s point in the same direction but offer different explanations. Crawford, Ezzell and Miles (1995) find that pay-performance sensitivity rose sharply after deregulation, particularly at less well-capitalized institutions, and read this as evidence of a moral hazard problem tied to mispriced deposit insurance. Hubbard and Palia (1995) confirm this direction using a much

larger and more finely dated sample, covering 147 banks with bank-specific deregulation dates, compared with the 37 banks and single regulated-versus-deregulated split used by Crawford et al. They also proposed another mechanism that believes that interstate deregulation has intensified competition for management talent: banks operating in more competitive markets need more capable CEOs, which requires higher and performance-related compensation to attract and retain them. The two explanations are not mutually exclusive, and neither study has completely ruled out the possibility of the other; on the contrary, the comparison shows that deregulation may increase the correlation between remuneration and performance through multiple channels at the same time.

Regarding tool selection, Chen, Steiner and Whyte (2006) found that after deregulation, banks are increasingly adopting option-based remuneration methods, and the remuneration structure and the accumulation of option wealth are positively related to risk-taking. The authors interpret this finding as evidence that management's risk-taking motivation outweighs its risk-avoidance motivation.

Competitive pressure continues to affect bank compensation after the initial impact of deregulation. A study by Cunat and Guadalupe (2009) shows that the intensity of competition in the banking and financial services industries has increased the compensation level of CEOs and their sensitivity to performance. Albuquerque, Cabral and Guedes (2019) further elaborated this view. They found that the same competitive pressure prompted banks to adopt more relative performance evaluations. This eliminates the impact of the overall impact of the industry on the personal remuneration of CEOs, but also makes management's investment choices tend to cross-bank correlation strategies: the incentive mechanism at the company level is more coordinated, but systemic risks increase. Overall, these findings have a clear revelation for financial stability. If competitive pressure prompts each bank to pursue stronger short-term incentives and a higher level of risk at the same time, even if each bank's own contract seems reasonable, the overall risk of the industry may exceed the risk level selected by any bank board.

Therefore, the level and structure of bank remuneration are not decided by the board of directors alone. They will be influenced by the regulatory and competitive environment around the bank, and to some extent constrained by these environments. Whether this model is also applicable to smaller institutions with less supervision remains to be studied empirically.

7.4 Empirical Evidence Linking Compensation Structure to Bank Risk-Taking

DeYoung, Peng and Yan (2013) used data from Commercial Bank of America from 1994 to 2006 to find that the higher the CEO's option compensation, the more subsequent risk-taking behavior, including higher leverage, greater profit fluctuations and a larger proportion of non-traditional business. Bai and Elyasiani (2013) found that the higher the Vega value, which measures the sensitivity of CEO wealth to stock earnings fluctuations, the higher the bank risk (measured by default distance), although this relationship is much weaker in well-governed banks. In other words, a strong incentive mechanism requires effective supervision to avoid excessive risks; the incentive mechanism itself does not explain all the problems.

The evidence provided by the crisis itself is more complicated about the relationship between the wage level and the salary structure. The study of Bhagat and Bolton (2014) found that when CEOs hold more stocks than options, the decline in bank's stock prices is smaller and less likely to need government support during the crisis, which is consistent with the view that restricted stocks are more likely to expose executives to downside risks than options. Salary levels and bonus payments alone do not predict worse crisis performance, which shows that composition is more important than size. The study of Fahlenbrach and Stulz (2011) complicated the situation. They found that bank executives who held more personal equity actually suffered greater personal wealth losses during the crisis, which meant that the mismatch of incentive mechanisms was not the whole reason. Many bank executives do seem to misjudge the risks faced by their institutions, which means that no matter how well-established the remuneration contract is, compensation design alone cannot replace sound risk management and prudent supervision.

Individual incentives connect to system-wide risk too. Albuquerque, Cabral and Guedes (2019) show that when investment opportunities are correlated across firms, risk-sharing between shareholders and managers produces compensation contracts built around relative performance evaluation, and that these contracts tilt investment toward correlated, systemically risky strategies, an effect that leverage makes worse. Policies aimed at individual banks' pay contracts, in other words, can be undone in aggregate by the very competitive and risk-sharing forces that shape the industry as a whole.

Table 7.1: Selected empirical studies on bank executive compensation and risk

Study	Methodology	Sample	Main findings	Limitations
Crawford, Ezzell and Miles (1995)	Pre/post comparison around US interstate deregulation	37 US banks	Pay-performance sensitivity rose after deregulation, more so at weakly capitalised banks	Small sample; single regulated/deregulated split
Hubbard and Palia (1995)	Pre/post comparison, bank-specific deregulation dates	147 US banks	Confirms rise in pay-performance sensitivity; attributes part of it to competition for talent	Deregulation timing not fully exogenous to bank characteristics
DeYoung, Peng and Yan (2013)	Panel regression	US commercial banks, 1994-2006	Higher CEO option pay linked to higher leverage, earnings volatility, non-traditional activities	Standard endogeneity concerns; options and risk jointly chosen
Bai and Elyasiani (2013)	Panel regression, distance-to-default	US bank holding companies	Higher vega linked to greater risk; effect weaker at well-governed banks	Governance quality itself hard to measure cleanly
Fahlenbrach and Stulz (2011)	Cross-sectional analysis of crisis outcomes	Large US banks, 2007-2008	CEOs with larger equity stakes suffered larger personal losses; incentive misalignment not the main story	Crisis period only; may not generalise to normal times
Beltratti and Stulz (2012)	Cross-country panel	Large banks worldwide	Shareholder-friendly governance associated with worse crisis performance	Crisis-period focus; governance measures vary by country
DeYoung and Huang (2021)	Panel regression, externality decomposition	Large US commercial banks, 1994-2016	Pay-performance incentives reduce both systemic-risk and liquidity-creation externalities	Liquidity-compensation link is a young literature; US-only sample

7.5 Corporate Governance of Banks and the Limits of Shareholder-Oriented Reform

Separating bank compensation design from the wider question of bank governance turns out to be impossible. Macey and O'Hara (2003) argue that a bank's principal claimants include depositors and other creditors alongside shareholders, so banks face a governance problem that standard mechanisms, such as independent boards and shareholder-aligned incentive plans, are poorly suited to. Those mechanisms were built to protect shareholders, not the wider set of stakeholders exposed to bank risk.

This concern is not just theoretical; the crisis itself confirms it. Beltratti and Stulz (2012) found: banks with shareholder-friendly governance actually performed worse during the crisis. This is because governance that serves shareholder interests encourages risk-taking, which benefits shareholders but harms creditors and depositors. Laeven and Levine (2009) also found similar results using transnational data: banks with concentrated equity and strong shareholder control are more risky. This model has been established in various national regulatory and governance systems, which excludes the possibility that this is only a unique phenomenon in the United States.

After investigating relevant literature, De Haan and Vlahu (2016) concluded that the role of traditional governance mechanisms, such as the composition and ownership structure of the board of directors, is systematically weak and unstable in banks for two reasons: first, regulation replaces some of the roles that were originally played by governance; second, the lack of transparency of bank balance sheets limits the achievements that even a well-intentioned board of directors can achieve. Taken together, this evidence pushes back against an assumption embedded in much of the general corporate-governance literature, that strengthening shareholder-oriented governance reliably improves outcomes. In banking, it may do the opposite. This is arguably the single most under-appreciated finding in the banking-compensation literature: standard governance mechanisms do not just fail to help, they can actively worsen outcomes.

7.6 Post-Crisis Regulatory Responses: Capital, Liquidity, Deferral and Clawback

In the post-crisis era, supervision has become the core corrective mechanism, which is not unreasonable. As mentioned earlier, competitive pressure and shareholder-oriented corporate governance have pushed the remuneration structure of banks to take more risks. Some jurisdictions have introduced rules that are directly binding on the remuneration structure. The European Union's Capital Requirements Directive IV requires that at least 50% of floating remuneration should be paid in the form of equity or equity-linked tools, and at least 40% of bonuses should be deferred for three to five years, and the ratio of floating remuneration to fixed remuneration must be capped. The British Prudential Regulation Authority and the Financial Conduct Authority have also introduced roughly similar deferred requirements and added recovery and penalty clauses, allowing banks to reduce or recover floating pay in the event of

misconduct or significant risk losses in the future (Angeli and Gitay, 2015). The scope and enforcement of these rules vary greatly in different jurisdictions, and their actual consistency is also questioned.

It is crucial to clarify how these three tools actually work, because they play a role at different stages of the pay cycle, and it is this time difference that prompts regulators to combine the three instead of relying on either tool alone. Deferred payment is the simplest of the three ways. It will not reduce the amount that executives can finally get, but only delay payment. The bonus is not paid in full at once at the end of the year, but part of it is paid immediately, and the rest is paid in installments over the following years. According to the regulations of the United Kingdom and the European Union, it is usually three to seven years (Angeli and Gitay, 2015). The deferred payment is for the regulatory authorities to gain time. It extends the time window for the consequences of risk decision-making, so that the relevant remuneration can be revealed before it is finalized and cannot be investigated.

Malus operates during this window period. Once the bonus has been earned, but before the deferred part has vested, if there is a loss, risk management error or misconduct, the bank may reduce or cancel the amount that has not yet vested. The bonus will not be paid in full or in part. However, malus is only valid during the deferral period. Once the bonus has vested and is paid, malus can no longer affect it, and this is where the third tool comes in.

Clawback aims to make up for this remaining loophole. If there is a serious problem afterwards, it allows the bank to recover variable remuneration that has already vested (in some cases, even paid in cash) (Angeli and Gitay, 2015). Among these three tools, clawback is the most powerful because it removes the protection that vesting would otherwise provide, but it is also the most difficult to implement. Recovering funds that executives have already spent, or that have moved beyond the reach of a given jurisdiction, raises legal and practical problems that neither deferral nor malus has to confront.

From this perspective, these three tools are not so much three independent options as a whole. Deferred payment provides a window period so that the risk results can be revealed. Malus allows the bank to take action within that window, before the remuneration is finally determined. Clawback extends the bank's power beyond that window, that is, to after the remuneration has

already been paid. If only deferred payment is implemented without malus or clawback, once a problem arises, the bank will have almost no actual ability to respond. That's why the mechanisms adopted by the European Union and the United Kingdom after the crisis bundled these three tools together instead of relying on any of them alone.

These remuneration-related rules are closely related to capital and liquidity regulation, rather than operating independently. Bhagat and Bolton (2013) believe that the requirement to hold a significant part of executive compensation in the form of restricted stocks and restricted stock options (the value of these instruments is directly related to bank losses) actually reinforces rather than replaces higher capital requirements, because both increase the cost of managers operating under insufficient capital buffers. Kolm, Laux and Loranth (2017) point out from a theoretical perspective that compensation supervision and capital supervision target different levels of risk transfer problems, and the combination of the two is best. Thanassoulis (2012) believes that the competition between banks for talents will weaken the unilateral constraints of any bank, so only the regulation of the remuneration structure across the economy can stop the prevalence of excessive risk.

Although the available evidence is of reference value, there is still insufficient evidence on whether these rules are really valid. Kleymenova and Tuna (2021) studied the introduction of the UK Remuneration Code, which mandated deferred compensation and performance-based vesting for bank executives, and find that UK banks' contribution and sensitivity to systemic risk rose by less than that of comparable UK non-bank firms after the reform, consistent with the expected effect of extending executives' decision-making horizon. However, they also document a rise in unforced CEO turnover at UK banks relative to other UK firms, suggesting the reform carried a cost in managerial retention. However, the rules for specific remuneration instruments are not inexplicable: the increase in fixed wages is to make up for the upper limit of bonuses, although fixed wages are more difficult to recover and the incentive effect is weaker. Some commentators worry that rigid compensation structure rules will only push talents to less regulated competitors or jurisdictions, which may undermine the goal of the regulation itself.

Therefore, solvency risk has always been the focus of the discussion of compensation supervision, which is not difficult to understand, because this is the channel where compensation has the most significant impact on the crisis. Liquidity risk received much less attention,

although it was equally crucial for the 2007-2009 crisis. These two risks are conceptually completely different. A bank's assets exceeding its liabilities on the books may still maintain its solvency, but will eventually go bankrupt because it cannot repay its short-term debts on time; this is exactly the case in several bankruptcy cases in 2008, including some institutions whose capital adequacy ratios were not low before the crisis broke out. Liquidity itself has been widely studied in a broader banking literature, mainly as an independent issue, related to deposit runs, wholesale financing, and the post-crisis Liquidity Coverage Ratio and Net Stable Funding Ratio. However, the relationship between bank liquidity and executive compensation rarely receives direct attention, which seems to be a real gap in the literature rather than a settled question.

A recent study has begun to fill this gap. DeYoung and Huang (2021) built a framework to link the compensation incentives of bank executives with two externalities that are not only borne by shareholders and have an impact on society: systemic risk and liquidity creation. Using data from large commercial banks in the United States from 1994 to 2016, they found that the pegged CEO's remuneration to performance would reduce negative externalities (systemic risk) and positive externalities (liquidity creation), while the linked pay to risk would increase both. Their core view is that these two externalities are interrelated rather than independent, so a rule aimed at reducing systemic risk through compensation may also reduce the liquidity that banks provide to the overall economy. This is a rare evidence that directly links compensation design to liquidity results, but so far it is almost the only such study.

This leaves an important practical question: how should the deferral, malus, and clawback mechanisms described above, aimed at dealing with capital loss and misconduct, interact with the liquidity situation of banks? At present, deferred remuneration is linked to the capital ratio and risk results (if there is any correlation), rather than liquidity indicators such as the Liquidity Coverage Ratio. Whether malus or clawback triggers should also reference liquidity pressure (not just solvency pressure or misconduct) has hardly been discussed at all in the literature reviewed in this article. Given that the liquidity crisis (not bankruptcy in the narrow sense of accounting) was the direct cause of several major bankruptcies in 2008, this shortcoming cannot be ignored. Supervision pays too much attention to individual compensation instruments and ignores the liquidity dimension, which may transfer rather than solve the fundamental risk-taking problem, because even if the bank's remuneration system successfully suppresses solvency risk,

the management may still lack the motivation to manage liquidity risk prudently.

7.7 Critical Assessment of the Banking Compensation Literature

Despite the substantial progress made since the crisis, there are still many limitations in the banking remuneration literature. A large number of empirical studies rely heavily on the crisis period itself, which is an extraordinary extreme event, and its lessons may not be extended to a calmer period. In the banking industry, it is also extremely difficult to establish a causal relationship, because bank risk, governance quality, remuneration design and supervision are all interrelated. The lack of research on changes in real extrinsic remuneration risk will eventually confuse the impact of remuneration with the impact of management quality, risk culture or regulatory system itself.

Most research is also concentrated on large listed banks in developed economies, while emerging markets, cooperative banks and savings institutions are relatively lacking in research. The decision-making of middle managers, traders and risk officers will have a substantial impact on the stability of the bank, but their remuneration has received far less attention than that of CEOs, and this imbalance has never been really corrected in the literature. The above discussion also reveals another obvious gap: there is much more knowledge of capital regulation than liquidity regulation in this field.

No remuneration design is the best in the banking industry. The appropriate balance between incentive mechanism and stability depends on the bank's business model, regulatory environment, ownership structure and the quality of internal risk management. In such a hugely different industry, unified and one-size-fits-all rules never work.

8. Connecting the Themes: Toward an Integrated View

8.1 The Common Thread: Incentive Design Under Imperfect Information

At first glance, the three themes of option incentives, cost-value gap and bank risk seem to be independent, but in fact they all face a common fundamental problem: the reason why incentive design is difficult is that the degree of effort of employees cannot be directly observed, executives generally aversion to risk, and the company's performance is subject to factors

beyond the control of managers.

Option compensation is an attempt to solve this problem. It can keep the interests of executives and shareholders consistent without anyone directly observing the efforts of employees. However, convex returns also bring their own problems: manipulation, short-termism, and high social cost risks in the banking industry. The cost-value gap reflects the same fundamental problem from another perspective. When companies try to provide attractive incentives, risk avoidance and investment diversification will weaken the actual value obtained by executives, so the final incentive effect is often lower than the level shown by the book data. Deposit insurance and limited liability system have especially amplified the inefficiency of the banking industry, making the social cost of taking risks far beyond what private contracts can afford.

8.2 The Governance Link

Corporate governance links these three themes. Good corporate governance helps the board of directors to find out whether the compensation incentive mechanism will have an adverse impact in time and make corresponding adjustments. As shown by cases such as option backdating, earnings management and "pay-for-luck", these problems are more prominent in companies with poor corporate governance (Core, Holthausen and Larcker, 1999; Garvey and Milbourn, 2006; Bebchuk, Grinstein and Peyer, 2010).

Corporate governance is particularly important in the banking industry, because external stakeholders such as regulators and deposit insurance institutions are directly related to the results of the banking industry. As Beltratti and Stulz (2012) pointed out, bank governance, which prioritizes shareholder influence, is associated with poorer crisis response performance, which denies simply copying the governance practices of non-financial enterprises. Bank governance must weigh the interests of creditors and the public, and give the same importance to the interests of shareholders, not replace the interests of shareholders. This is exactly the intersection of the optimal contract theory and the rent-extraction theory camp. Both sides agree that the quality of governance will affect the remuneration results. The only difference is whether the observed practice effectively adapts to governance constraints or takes advantage of governance defects. The evidence reviewed in this article shows that both mechanisms are working.

8.3 Accounting Standards as a Structural Determinant

The incentive mechanism is influenced by accounting standards, and its influence is no less than anyone's intentional design. The rise and fall of U.S. option compensation clearly shows this: the rise and fall of stock options is more closely related to changes in accounting standards than changes in contract theory or management preferences.

Accounting costs and perceived value are directly related to a broader literature on incentive theory. When the enterprise partially chooses remuneration tools for accounting considerations, the observed remuneration structure reflects the comprehensive results of accounting treatment, tax rules and regulatory constraints, rather than the pure optimal incentive design. The empirical study of the observed remuneration as the best contract result may unconsciously measure completely different factors.

Applied to the banking industry, regulatory requirements such as deferred remuneration and penalty clauses may also play a role in part through accounting channels. When recoverable deferred remuneration and immediately effective remuneration differ in the processing of financial reports, the enterprise's response to accounting processing is as important as the response to the potential incentive logic, and most regulatory design literature ignores this subtlety.

8.4 Toward More Integrated Research

These factors deserve to be studied comprehensively, rather than viewed in isolation, which can directly lead to several problems. How does the cost-value gap affect the results of corporate governance? When executives have significant discount options, the actual incentive effect will be lower than the level shown by accounting costs, and the board of directors who rely on these figures will eventually overestimate the enthusiasm of executives unconsciously. Research on measuring perceived value rather than accounting costs may reveal a stronger and more consistent link between compensation and behavior than existing literature.

Does the complexity of the contract itself constitute a corporate governance problem? Literature shows that complex performance equity will bring valuation challenges and reduce transparency, while transparency can reduce opportunistic behavior; however, even under strong corporate

governance, there is still no conclusion on whether the complexity of compensation design will lead to rent-extraction behavior.

How should the unique governance mechanism of banks interact with the remuneration mechanism? Traditional corporate governance reforms cannot directly improve the stability of banks. Therefore, the remuneration design of banks needs to balance the regulatory framework, capital and liquidity structure, and the interests of non-shareholder claimants at the same time. This is more complicated than any design problems faced by non-financial enterprises.

9. Conclusion and Future Research Directions

This review first discusses two related issues in the executive compensation literature, and then introduces the banking industry as a third perspective to make these two issues more visible. The first question is behavioral: how does option compensation affect the degree of effort and risk-taking of executives? The second question is at the measurement level: the gap between the company's book compensation cost and the economic value actually perceived by executives. The banking industry highlights these two problems, because the consequences of the misalignment of the incentive mechanism are not limited to shareholders, but also to depositors and creditors, and ultimately affect taxpayers.

In terms of executive behavior, options do have a measurable impact. Risk-taking will increase with the increase of the Vega value, and the degree of effort will also respond to the equity ratio, although the evidence of the degree of effort is not as strong as the evidence of risk-taking, because the degree of effort itself is difficult to observe directly. Options also bring significant side effects: earnings management, opportunistic timing, short-termism, and excessive preference for systemic risks in the banking industry. These impacts largely depend on the specific situation and corporate governance. When corporate governance is weak or risk-taking brings costs to external parties of the company, these effects cause the greatest damage, and the banking industry is structurally responsible for this situation.

In terms of cost value, accounting-based measures may substantially exceed the economic value that executives privately assign to equity awards, particularly when executives are risk-averse and poorly diversified. The Black-Scholes model is normally used to estimate market or fair

value for accounting purposes, not the certainty-equivalent value that executives themselves assign to their awards. Early exercise means that the same nominal compensation scheme has different value for different executives, depending on the factors that are never reflected in the accounting report. This empirical study of mismatched remuneration design, benchmarking practice and compensation performance sensitivity are all of practical significance, and its importance may exceed the importance of existing literature, because many studies still use the reported remuneration cost directly as an indicator of incentive intensity.

These two problems are particularly prominent in the banking industry. The convex remuneration structure encourages risk-taking. Although this is beneficial to shareholders, it also exposes creditors, depositors and the general public to losses. Regulatory measures, including deferred pay, malus and clawback provisions, as well as bonus ceilings linked to capital and liquidity, have had a certain effect, but the effect is limited. Some evidence shows that regulatory measures also bring unexpected consequences. For example, after the bonus ceiling is set, the fixed salary rises, which quietly raises another problem while the regulation solves one problem. The quality of corporate governance is the main line that runs through all three areas: good corporate governance can mitigate the negative impact of incentive compensation, while weak corporate governance will aggravate these effects. This point matters more than any single policy lever, and it is also the conclusion repeatedly emphasized in this review.

These findings are of practical significance beyond academic literature and directly affect the actions of the board of directors, the compensation committee and regulators. The most core lesson is that the remuneration structure is more important than the remuneration level. Therefore, the committee that formulates executive remuneration should focus on the incentive structure rather than the total remuneration. Among the existing design methods, the vesting period is the most critical, because a longer vesting period can reduce short-termism and strengthen consistency with long-term value creation; British corporate governance guidelines have advocated the requirement to hold options after retirement and have been supported by theoretical literature (Edmans et al., 2012), which further strengthens this point, although in practice, its implementation is still far behind the relevant theoretical publicity. The cost-value gap is not only reflected in the contract design, but also in the compensation benchmarking. The remuneration committee should try to evaluate the perception of the actual value of their

compensation plans by executives of peer companies, rather than simply comparing the accounting costs of each company. Although proxy consultants and compensation consultants can use the vesting schedule and option value data to estimate perceived value, this work is very difficult in practice. Although this is not perfect, it is better than treating accounting figures as real numbers.

Specifically for the banking industry, its actual impact is more significant, because the stakes are far more than the company itself. In particular, regulators and bank boards have good reason to combine multiple compensation tools instead of relying on any single one. Internal debts, deferred remuneration, and pensions recovered when the risk outcome worsens can align executives with the interests of creditors and deposit insurance funds, thus reducing the moral hazard of limited liability. As mentioned in Chapter 7, remuneration supervision and capital supervision target different aspects of risk transfer. Therefore, these measures should coexist with capital and liquidity requirements, rather than replace them. Moreover, since deferral, malus and clawback currently depend almost entirely on the trigger mechanism of capital and misconduct, given the key role played by the liquidity crisis in the 2008 financial crisis, it is best for regulators to clearly ask whether liquidity pressures (not just solvency pressures) should also trigger malus or clawback provisions when reviewing compensation policies. All of these cannot replace the need for direct prudential supervision. A well-designed compensation system can at best reduce the negative impact of compensation on prudential goals, but it can never replace prudential goals themselves.

The shortcomings found in this review lead to several directions for future research. The first is the measure: if the study uses the perceived value of executives instead of accounting costs as the measure of remuneration, it will help to identify the real incentive effect more accurately. This requires collecting first-hand data directly from executives or building a better structural model to describe how risk-averse entities with non-diversified portfolios evaluate compensation instruments that they cannot trade freely. This remains an underdeveloped field in the literature.

Next is the scope of research. Due to the huge differences in institutional environment, effective methods for listed companies in the United States may not be applicable to other regions, so more research is needed in managers outside the United States, private enterprises and below the CEO level. Transnational research linking compensation structure with financial stability is

particularly important for the banking industry.

The third and most specific shortcoming found in this review is the liquidity problem. Compared with capital supervision, the research on how liquidity regulation interacts with remuneration design is still relatively scarce. DeYoung and Huang (2021)'s research on the externality of liquidity creation is one of the few starting points in this field at present. Some promising expansion studies include a direct examination of whether deferral length, malus triggers, or clawback conditions related to liquidity indicators such as the Liquidity Coverage Ratio or the Net Stable Funding Ratio have a different impact on banks' liquidity creation behavior than remuneration instruments that are only linked to capital or solvency indicators. Since this problem is at the intersection of two main independent development areas, remuneration research and liquidity research, it may require cooperation between sub-fields that currently have few exchanges with each other.

The fourth question involves corporate governance: how to interact with other corporate governance mechanisms in remuneration design in practice. As we all know, the quality of corporate governance will regulate the impact of remuneration, but little is known about the channels in which it plays a role. The study of combining specific corporate governance interventions with the precise measurement of remuneration adjustment helps to clarify these mechanisms.

The fifth problem, perhaps the most difficult problem to solve, is dynamism. Most studies rely on a shorter time window and regard compensation as static. In fact, contracts will evolve with changes in company performance, CEO tenure and market conditions. The dynamic contract model can predict how the incentive mechanism will change over time, but these predictions have only been partially verified at present. Longer panel data sets and more accurate causal identification are not optional extras that can be dispensed with. Otherwise, the existing literature will continue to answer questions about compensation snapshots, rather than questions about how compensation actually plays a role in executive careers.

In a word, it is difficult to draw a simple conclusion about executive compensation from the evidence examined in this review. Options do affect behavior. Accounting-based measures can give an incomplete or potentially misleading picture of incentive intensity. These problems can

be amplified in banking because of leverage, deposit insurance and systemic externalities. The existing literature provides not a single conclusion, but a series of design principles: extending the research cycle, treating accounting costs as approximations rather than facts, combining remuneration tools with capital and liquidity supervision, rather than expecting any single tool to do everything, and keeping in mind that the quality of corporate governance determines the effectiveness of these measures in practice. Building a literature system that can directly test these principles, instead of continuing to infer these principles from accounting agent data and crisis data, is the task left by this review for future research.

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